The Normal Distribution

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.03

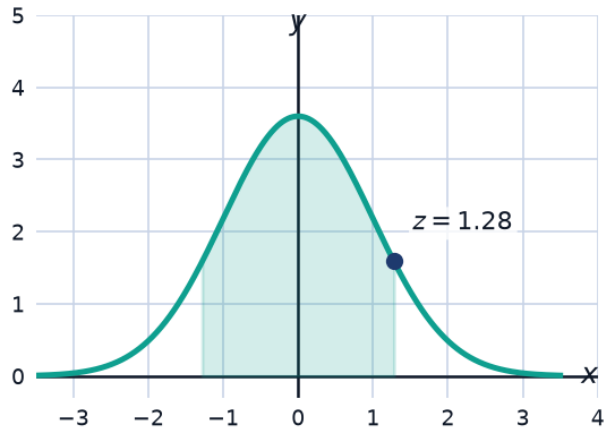
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Properties of the normal distribution

State three key properties of a normal distribution's shape.

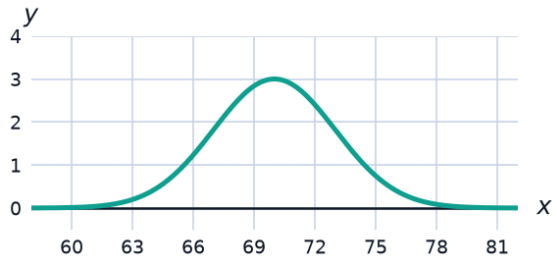
2 [Foundational] • Target: Properties of the normal distribution

Explain what the two parameters μ and σ control in a normal distribution's shape.



3 [Foundational] • Target: The 68-95-99.7 rule

Adult male heights are approximately normal with mean 70 inches and standard deviation 3 inches. Using the 68-95-99.7 rule, find the range of heights containing the middle 95% of adult men.



4 [Foundational] • Target: The 68-95-99.7 rule

Using the same distribution, find the approximate percentage of adult men taller than 76 inches.

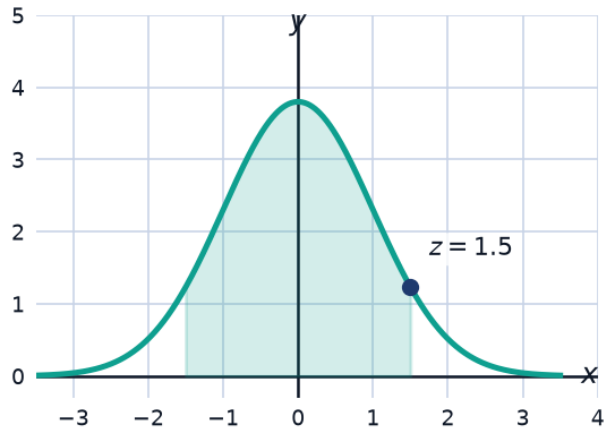
5 [Foundational] • Target: Standardizing and using the normal distribution

Using the same height distribution (mean 70, standard deviation 3), find the z-score for a height of 77.5 inches, and use a standard normal table value of $P(Z < 2.5) \approx 0.9938$ to find the proportion of men shorter than 77.5 inches.

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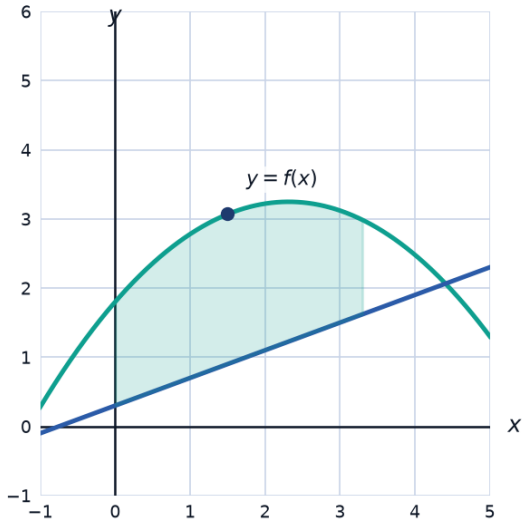
6 [Foundational] • Target: Standardizing and using the normal distribution

Find the proportion of men between 65 and 75 inches tall, given $P(Z < 1.67) \approx 0.9525$ and $P(Z < -1.67) \approx 0.0475$.



7 [Foundational] • Target: Visualizing the standard normal distribution

The graph shows the standard normal distribution ($\mu = 0$, $\sigma = 1$), with the area to the left of $z = 1$ shaded (though shading itself is not shown, imagine it visually).



a Using $P(Z < 1) \approx 0.8413$, find the area under the curve to the right of $z = 1$.

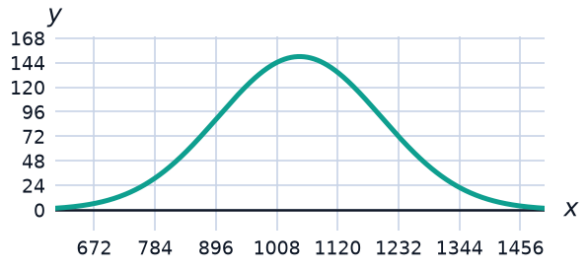
b Explain, using the curve's symmetry, why $P(Z < -1) = P(Z > 1)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Working backward: finding values from percentiles

SAT scores are approximately normal with mean 1050 and standard deviation 150. A student scores at the 90th percentile, corresponding to $z \approx 1.28$. Find that student's approximate score.

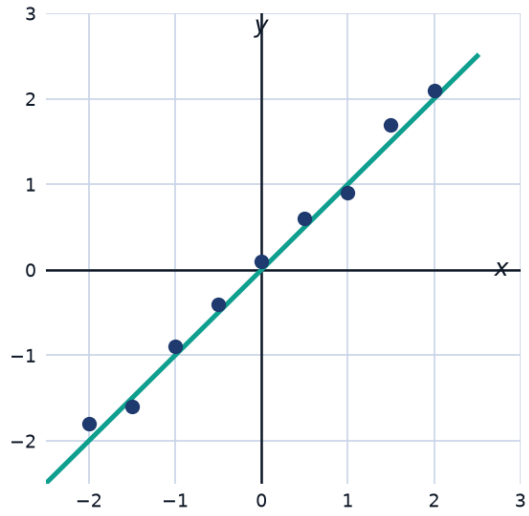


9 [Intermediate] • Target: Working backward: finding values from percentiles

Using the same distribution, find the score that separates the bottom 10% of scores, given that $z \approx -1.28$ corresponds to the 10th percentile.

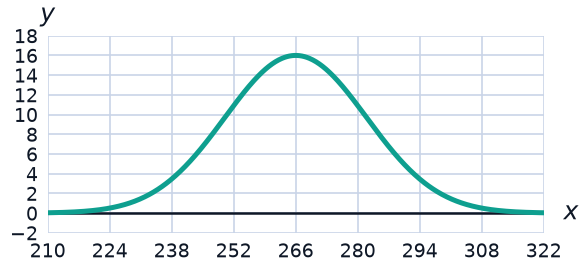
10 [Intermediate] • Target: Assessing normality

Explain how a normal probability plot (or normal quantile plot) can be used to assess whether a data set is approximately normal, and what a roughly straight-line pattern indicates.



11 [Intermediate] • Target: Applying the normal distribution to real data

Pregnancy lengths are approximately normal with mean $\mu = 266$ days and standard deviation $\sigma = 16$ days. Find the z-score for a pregnancy lasting 250 days.



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12 [Intermediate] • Target: Applying the normal distribution to real data

Using the same pregnancy-length distribution and the 68-95-99.7 rule, find the approximate percentage of pregnancies lasting between 234 and 298 days.

13 [Intermediate] • Target: Combining two independent normal random variables

A vending machine dispenses a soda whose volume is normally distributed with mean 355 mL and standard deviation 4 mL, independently of a snack bag whose weight is normally distributed with mean 50 g and standard deviation 2 g. If T represents the combined "total" formed by adding the soda volume (in mL) and snack weight (in g) as a single number, find the mean of T. (This is a simplified illustrative combination, not a real physical unit.)

14 [Intermediate] • Target: Combining two independent normal random variables

Battery lifespans are normally distributed with mean 500 hours and standard deviation 40 hours. Find the z-score for a battery lasting 460 hours, and describe its relative position.

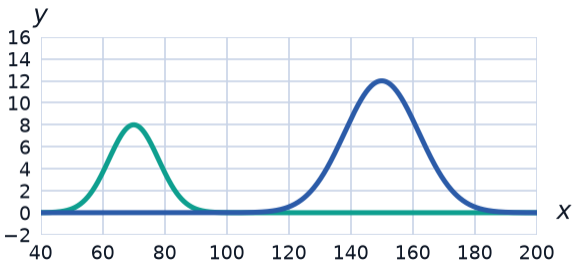
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Combining two independent normal random variables • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

A company manufactures resistors with resistance values approximately normal, mean 100 ohms and standard deviation 5 ohms. A resistor is considered defective if its resistance is below 90 ohms or above 110 ohms. Using the 68-95-99.7 rule, estimate the percentage of resistors that are defective, and explain your reasoning step by step.

16 [Challenging] • Target: Combining two independent normal random variables • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

Two students are comparing their standardized test scores from different tests. Student X scored 78 on a test with mean 70 and standard deviation 8. Student Y scored 165 on a different test with mean 150 and standard deviation 12. Using z-scores, determine who performed relatively better compared to their own test's distribution of test-takers, showing full reasoning.



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17 [Challenging] • Target: Properties of the normal distribution • Cross-Lesson: 2.01 Sampling Methods & Bias

The distribution of cholesterol levels in a population is approximately normal with $\mu = 190$ mg/dL and $\sigma = 24$ mg/dL. (a) What proportion of individuals have cholesterol levels above 226 mg/dL? (b) What cholesterol level marks the 90th percentile?

18 [Challenging] • Target: The 68-95-99.7 rule • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

A basketball player makes 75% of free throws. In a game, she shoots 12 free throws. Assuming attempts are independent: (a) Find the probability she makes exactly 9 free throws. (b) Find the mean and standard deviation of made free throws.

19 [Challenging] • Target: Standardizing and using the normal distribution • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

[Unit 1 · Exploring One-Variable & Two-Variable Data · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

20 [Challenging] • Target: Visualizing the standard normal distribution • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

Two independent random variables are normally distributed as $X \sim N(50, 16)$ and $Y \sim N(45, 9)$. Find the probability that $X > Y$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Properties of the normal distribution

A normal distribution has mean 64 and standard deviation 9. Using the 68-95-99.7 rule, find the range containing the middle 95% of values.

a. $(46, 82)$

b. $(37, 91)$

c. $(46, 81)$

d. $(47, 82)$

22 [Foundational] • Target: The 68-95-99.7 rule

A normal distribution has mean $\mu = 101$ and standard deviation $\sigma = 4$. Find the approximate percentage of values below $x = 103$, to two decimal places.

a. 69.15%

b. 72.15%

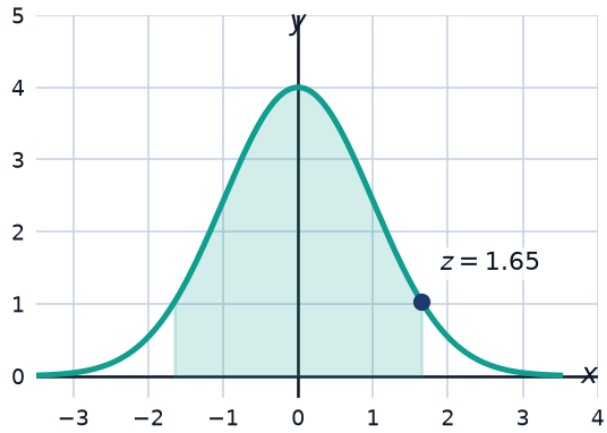
c. 66.15%

d. 30.85%

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23 [Foundational] • Target: Standardizing and using the normal distribution

A normal distribution has mean $\mu = 144$ and standard deviation $\sigma = 16$. Find the approximate percentage of values between $x_1 = 136$ and $x_2 = 152$, to two decimal places.



a. 69.15%

b. 34.29%

c. 42.29%

d. 38.29%

24 [Foundational] • Target: Visualizing the standard normal distribution

Scores on an exam are approximately normal with mean $\mu = 390$ and standard deviation $\sigma = 15$. A student scores at the 95th percentile, corresponding to $z \approx 1.645$. Find that student's approximate score.

a. 365

b. 405

c. 425

d. 415

25 [Foundational] • Target: Working backward: finding values from percentiles

A company manufactures parts with a measurement approximately normal, mean 152 and standard deviation 5. A part is considered defective if its measurement is below 142 or above 162 (i.e. outside $\mu \pm 2\sigma$). Using the 68-95-99.7 rule, estimate the percentage of parts that are defective.

a. 0.3%

b. 5%

c. 32%

d. 2.5%

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26 [Foundational] • Target: Assessing normality

A data set has $Q_1 = 17$ and $Q_3 = 35$. Using the 1.5×IQR rule, which of the following values would be classified as an outlier?

a. 35

b. -14

c. 17

d. -9

27 [Foundational] • Target: Applying the normal distribution to real data

A researcher is designing an experiment to test the effectiveness of a new fertilizer on plant growth. They have a total of 50 experimental plots available. The experiment will involve two treatments: the new fertilizer (Treatment A) and a standard fertilizer (Treatment B). To ensure fair comparison and control for confounding variables, they decide to use a randomized block design. The plots are divided into 5 blocks based on soil quality, with each block containing 10 plots. Within each block, the plots will be randomly assigned to either Treatment A or Treatment B. If the number of plots assigned to Treatment A within the first block is x, and the total number of plots assigned to Treatment A across all 5 blocks is 25, and the proportion of plots assigned to Treatment A in block 1 is $\frac{1}{5}$ of the total plots in block 1, what is the value of x?

a. 3

b. 4

c. 5

d. 6

28 [Foundational] • Target: Combining two independent normal random variables

Two independent random samples, each of size n, are taken from a normal population with mean μ and known standard deviation σ. From each sample, a symmetric 90% confidence interval for μ is constructed. Given that $\Phi(1.645) = 0.9500$ and $\Phi(2.326) = 0.9900$, where Φ denotes the standard normal cumulative distribution function, find the probability that these two confidence intervals overlap.

a. 0.980

b. 0.900

c. 0.810

d. 0.990

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Properties of the normal distribution

A researcher is designing a survey to understand student opinions on campus dining. The researcher decides to select students from the student directory using a systematic random sampling method. If the student directory has a total of 1500 students, and the researcher decides to select every 10th student, starting with a randomly selected student between 1 and 10, how many students will be in the sample?

a. 150

b. 10

c. 15

d. 1500

30 [Intermediate] • Target: The 68-95-99.7 rule

A data set contains the values 4, 13, 14, 23, 30, 33, 35. What is the mean of this data set?

a. 23

b. 21.71

c. 23.89

d. 19.71

31 [Intermediate] • Target: Standardizing and using the normal distribution

A data set contains the values 22, 25, 41, 51, 54, 64. What is the mean of this data set (rounded to 2 decimal places)?

a. 48.83

b. 38.83

c. 42.83

d. 45.83

32 [Intermediate] • Target: Visualizing the standard normal distribution

A researcher wants to investigate the relationship between the type of pet owned (Dog, Cat, Fish) and the preferred music genre (Rock, Pop, Classical). They collect data and present it in a two-way table. If the table has 3 rows and 3 columns, what is the total number of cells in the table that represent a combination of pet ownership and music genre?

a. 6

b. 7

c. 9

d. 16

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33 [Intermediate] • Target: Working backward: finding values from percentiles

A researcher is investigating the relationship between study habits and exam performance. They categorize students into two groups: 'Regular Study' and 'Cramming'. They also categorize exam performance into 'Pass' and 'Fail'. The data collected is presented in a two-way table. If the table has 2 rows representing study habits and 3 columns representing exam performance categories, what is the total number of cells in this two-way table?

a. 4

b. 5

c. 6

d. 3

34 [Intermediate] • Target: Assessing normality

The IQR (interquartile range) equals:

a. $Q_3 - Q_1$

b. $Q_1 - Q_3$

c. Max – Min

d. $Q_2 - Q_1$

35 [Intermediate] • Target: Applying the normal distribution to real data

From a randomly ordered list of 1854 customers, a researcher selects every 21th person after choosing a random starting point. What sampling method is this?

a. Convenience sample

b. Systematic sample

c. Stratified random sample

d. Simple random sample (SRS)

36 [Intermediate] • Target: Combining two independent normal random variables

From a randomly ordered list of 1002 employees, a researcher selects every 14th person after choosing a random starting point. What sampling method is this?

a. Cluster sample

b. Systematic sample

c. Voluntary response sample

d. Stratified random sample

37 [Intermediate] • Target: Properties of the normal distribution

From a randomly ordered list of 885 voters, a researcher selects every 24th person after choosing a random starting point. What sampling method is this?

a. Simple random sample (SRS)

b. Voluntary response sample

c. Convenience sample

d. Systematic sample

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The 68-95-99.7 rule • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

From a randomly ordered list of 610 students, a researcher selects every 18th person after choosing a random starting point. What sampling method is this?

a. Simple random sample (SRS)

b. Systematic sample

c. Convenience sample

d. Stratified random sample

39 [Challenging] • Target: Standardizing and using the normal distribution • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

From a randomly ordered list of 1390 athletes, a researcher selects every 7th person after choosing a random starting point. What sampling method is this?

a. Stratified random sample

b. Systematic sample

c. Simple random sample (SRS)

d. Voluntary response sample

40 [Challenging] • Target: Visualizing the standard normal distribution • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

From a randomly ordered list of 1515 patients, a researcher selects every 6th person after choosing a random starting point. What sampling method is this?

a. Convenience sample

b. Systematic sample

c. Simple random sample (SRS)

d. Stratified random sample

41 [Challenging] • Target: Working backward: finding values from percentiles • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

A data set (already sorted, n=9) is 1, 7, 10, 11, 25, 34, 43, 46, 51. What is the five-number summary?

a. Min = 1, Q1 = 10, Median = 51, Q3 = 43, Max = 25

b. Min = 1, Q1 = 43, Median = 25, Q3 = 10, Max = 51

c. Min = 1, Q1 = 10, Median = 25, Q3 = 43, Max = 51

d. Min = 51, Q1 = 10, Median = 25, Q3 = 43, Max = 1

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42 [Challenging] • Target: Assessing normality • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

An outlier in a boxplot is typically beyond:

a. 1.5 × IQR from Q1 or Q3

b. 2 × IQR

c. 3 standard deviations

d. The median

43 [Challenging] • Target: Applying the normal distribution to real data • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

A regression line predicts $\hat{y} = 41.85$ for $x = 25$, while the observed value is $y = 39.1$. What is the residual for this point?

a. 3.25

b. -2.75

c. -0.75

d. -6.75

44 [Challenging] • Target: Combining two independent normal random variables • Cross-Lesson: 1.01 Measures of Center & Spread

Which test compares two independent population means?

a. One-sample z-test

b. Paired t-test

c. Two-sample t-test

d. Chi-square test

45 [Challenging] • Target: Properties of the normal distribution • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

A randomised block design blocks subjects by age group into 3 blocks of 44 subjects each, with random assignment to treatments within each block. What is the total sample size?

a. 117

b. 107

c. 142

d. 132

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